

Disentangling Contextual Coherence and Prompt Echo in Early R-Lens Readouts

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CAMBRIA capstone — a blinded two-model independent evaluation and methodological extension of the R-Lens early-layer coherence claim

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Abstract

R-Lens uses the J-Lens fitting pipeline but replaces ordinary-gradient transport with LRP-derived relevance coefficients, applying modified backward rules for normalisation and gated MLP components while leaving the forward pass unchanged. Its authors report, qualitatively and without a released scorer, that R-Lens readouts at early layers are more coherent than J-Lens readouts. We evaluate that claim on two 27B models with a blinded, pre-specified and frozen rating panel of 200 three-arm cells, scored by two independently validated LLM autoraters.

Under a standard prompt-aware coherence rubric, R-Lens is rated more coherent than J-Lens on both models: +0.690, 95% CI [0.520, 0.855] on Qwen3.5-27B and +1.010, [0.785, 1.235] on Gemma-3-27B-it, on a 0–4 scale. R-Lens also surfaces significantly more prompt-overlapping content than J-Lens.

We therefore attempted, post hoc, to measure coherence with prompt-local content excluded: a second rubric instructing judges to discard tokens recoverable from the prompt and score only what remains. Under it the pooled R–J difference falls from +0.690 to +0.325 [0.165, 0.480] on Qwen and from +1.010 to +0.110 [−0.045, 0.255] on Gemma.

That second instrument did not validate, and we report it as an exploratory measurement attempt rather than a second endpoint. Its inter-judge agreement is slight ($\kappa = 0.19$ against 0.51); on Gemma its two judges disagree in sign with disjoint intervals; and in a mixture-invariance control holding a fixed coherent core while varying only the ignorable prompt-local padding, one judge held steady and the other’s scores declined monotonically with padding in half of all families. Excluding prompt-local content therefore did not produce a reliable cross-judge measure. That is itself a result: the decomposition of “coherence” into prompt-local and non-prompt-local components is measurement-sensitive.

The contribution is therefore neither a confirmation nor a refutation. R-Lens is rated more coherent under a standard rubric, robustly and on both models. It also receives more credit for prompt-local content — which includes orthographic normalisation such as *poeples* → *people*, plausibly genuine computation rather than copying. A single hand-selected cell shows the two rubrics ranking the same three arms in opposite orders, which demonstrates the construct distinction without needing any instrument to be reliable. We also show that restricting to cells with equal observed echo — the natural first response to this confound — gives a markedly more reassuring answer than rubric-based exclusion, because it conditions on a variable the lens itself affects. Neither method cleanly identifies the contribution of prompt-local content.

1 What this document does and does not claim

This is an independent evaluation of the coherence claim, and a methodological extension of it. It is *not* an exact replication: the source post released no scorer, so the rubric, panel and statistics here are ours, and the released lens artifacts we evaluate were fit with a lighter recipe than the one the post describes (§2).

It asks two questions. First, when a rater is shown three unlabelled early-layer readouts of the same prompt — one from the logit lens, one from J-Lens, one from R-Lens — does the R-Lens arm receive a higher contextual-coherence score? Second, does any advantage survive when the rater is forbidden from crediting content that is *prompt-local* — direct repetition together with trivial and normalised lexical variants?

It makes **no** claim about concept specificity, causal onset, ablation, necessity, sufficiency, answer smuggling, or whether any lens recovers a representation the model actually uses. Separate experiments in this project address some of those; by pre-specification their results are not pooled with, used to interpret, or reported alongside the coherence estimand, and they do not appear here.

Terminology. The protocol was written, frozen and hashed before any rating was collected, but it was not deposited with a third-party registry. We therefore say *pre-specified and frozen* throughout, never *preregistered*.

The two stages were not specified at the same time. The standard-rubric panel, rubric and analysis were frozen before any rating. The non-echo analysis was *not*: it was motivated after inspecting the prompt-echo result from the standard rubric. Its rubric, salt, controls and analysis procedure were then frozen and hashed before any non-echo rating was collected, and its copy control was run and passed before the panel was scored. It is therefore a *staged methodological extension*, not part of the original pre-specified endpoint, and readers should weight it accordingly. The same applies to the small-sample analyses in §7 and the qualitative examples in §6, both of which are post-hoc.

2 Artifacts and provenance

Both lenses are the authors’ released artifacts, not refits, taken at a pinned commit. A preflight check refuses to run if either lens fails provenance:

- J-Lens carries {"estimator": "standard"}.
- R-Lens carries {"estimator": "relp"} with the LN rule, identity rule and half rule at $\beta = 0.5$.
- **Both were fit with n_prompts: 25**—the released light recipe, not the $n = 1000$ recipe described in the source post. Any difference between our effect size and theirs is confounded with this.
- The two Jacobian sets are verified to actually differ, so a null could not be produced by silently loading the same lens twice.

Preflight additionally verifies hook indexing (that recorded block outputs satisfy `activations[ℓ]` equals `hidden_states[ℓ+1]`), target layer identity, and transport orientation. A failure at any of these is fail-closed: the run aborts rather than proceeding with a warning.

3 Method

3.1 Panel

Twenty prompts—four from each of the five official evaluation sets (association, multihop, multilingual, poetry, typo)—were selected by a hash of a fixed salt, the set name and a stable item identifier, so selection is reproducible and independent of any result. For the two sets carrying a

target (multihop, multilingual) only items the model itself answers correctly were eligible, matching the source protocol’s correctness filter.

The eligible pool is the intersection $\mathcal{E}_{\text{Qwen}} \cap \mathcal{E}_{\text{Gemma}}$: an item is eligible only if *both* models answer it correctly, and the four prompts per set are drawn from that shared pool by hash. Both models are therefore read on the same 20 items, and no item enters for one model on the strength of the other’s correctness alone.

Each prompt is read at five pre-specified relative depths $z = \ell/\ell^* \in \{0, 0.1, 0.2, 0.3, 0.4\}$, giving $20 \times 5 \times 2 = 200$ cells, each holding three arms. Readout position follows the source protocol: the final prompt token, except poetry, read at the newline ending line one.

3.2 Blinding

Arm order is permuted per (*cell*, *rater*) pair, not once per cell. A single fixed order would make position bias identical across judges and therefore invisible to any agreement statistic. The unblinding key was written outside the repository and outside the published result tree; the panel and the sample carry content hashes; and every outgoing judge payload was reconstructed after the fact and scanned for structural leakage of lens identity.

3.3 Rubric

Three dimensions per arm: **contextual coherence** (0–4, primary), **lexical integrity** (0–2) and **prompt echo** (0–2). Contextual coherence asks whether the readout tokens form a plausible continuation or elaboration of the prompt’s content. Prompt echo measures how much of the readout is copied from the prompt — it is scored so that copying can be detected, not rewarded.

3.4 Judges

Two primary autoraters, `openai/gpt-5` and `deepseek/deepseek-chat-v3.1`, each independently validated on a 78-cell battery of control categories before being admitted. Validation gates include position-bias rate (tested with an exact binomial, so that a threshold finer than the achievable resolution at that sample size returns UNDERPOWERED rather than a false pass), a corruption-detection control, a duplicate-arm control, and a response-completeness gate that fails a judge returning empty content on more than 2% of cells. One candidate judge was rejected under exactly this last gate.

The primary score is the **mean of the two validated judges**. A third judge (`meta-llama/llama-3.1-70b-instruct`) was originally used to adjudicate the 63% of cells on which the primaries disagreed, producing a median-of-three score. *That adjudicator did not clear validation*, and the primary estimate was therefore demoted to the mean of two before any result was written up. Section 5.2 gives the gate it failed to resolve, and Section 5.3 reports the adjudicated estimate as a sensitivity analysis.

This change increases the reported effect (pooled $+0.778 \rightarrow +0.850$), and we state that explicitly because a reader is entitled to ask whether the rule was chosen for its result. It was not: the demotion follows from an instrument failing its admission gate, the gate was specified before the adjudicator was run, and the direction and significance of every contrast are unchanged either way. Both estimates appear in this document.

3.5 Statistics

The estimand is the equal-weight paired difference: mean over evaluation sets, then prompts within set, then depths within prompt, so that no set or prompt is over-weighted by having more cells. Intervals are percentile 95% from a 10,000 replicate stratified paired *prompt-cluster* bootstrap. p -values are paired prompt-cluster sign-flip permutation tests — never a bootstrap sign proportion. Depth and set breakdowns are Holm-corrected within model.

What the intervals cover. The bootstrap resamples *prompts*, so every interval here quantifies variation across prompts *conditional on the rubric and the two judges we used*. It does not represent uncertainty over the population of possible evaluators. With two judges a random-effects model over raters is not available, so judge disagreement is reported directly — per-judge estimates side by side — rather than folded into a width.

4 Results

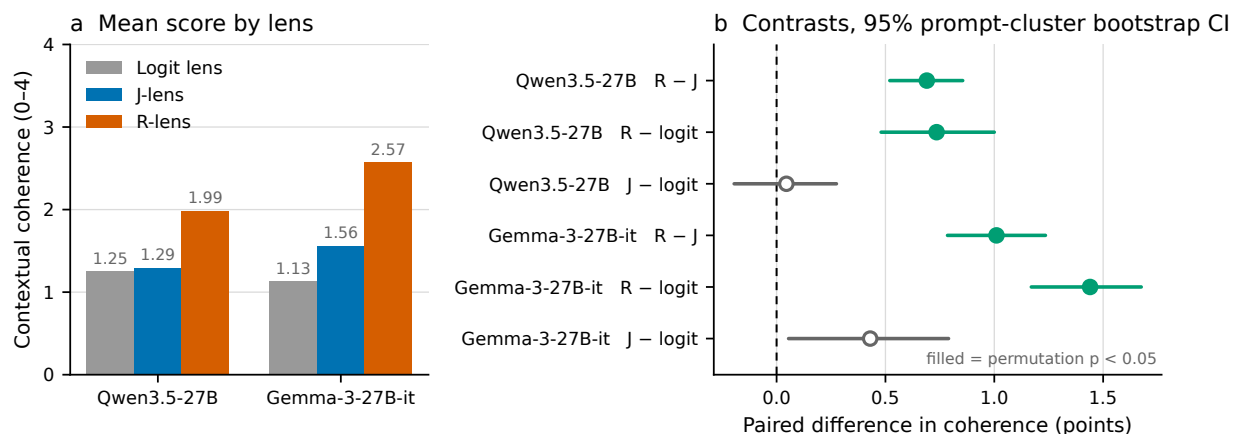


Figure 1: Primary result. (a) Mean contextual coherence by lens on the full 0–4 rubric scale. (b) Paired contrasts with 95% prompt-cluster bootstrap intervals; a filled marker denotes significance under the *pre-specified* sign-flip permutation test, which is not always the same as the interval excluding zero — J – logit on Gemma is drawn hollow for that reason. The axis in (a) is deliberately not rescaled: the ordering is clear, but every lens sits in the lower half of the scale.

4.1 Primary endpoint

R-Lens is rated more contextually coherent than J-Lens on both models (Table 1). Both intervals exclude zero in the positive direction, so the ordering holds on both models under this rubric.

Two features of this table matter as much as the headline.

J-Lens is not distinguishable from the logit lens on either model ($+0.045$, $p = 0.776$ on Qwen; $+0.430$, $p = 0.112$ on Gemma). On Qwen the two are almost exactly equal in mean score, 1.295 against 1.250 on a 0–4 scale. Under the primary rule the entire measured advantage over the logit-lens baseline is carried by the R-Lens variant, not by the Jacobian transport as such. Under the adjudicated rule the Qwen contrast is instead positive and significant, and Section 5.3 shows it is judge-unstable in sign; it should not be quoted as a single number in either direction.

Model	Contrast	Δ	95% CI	p	win	tie	loss
Qwen3.5-27B	R – J	0.690	[0.520, 0.855]	$< 10^{-4}$	.69	.20	.11
Qwen3.5-27B	R – logit	0.735	[0.480, 1.000]	$< 10^{-4}$	.64	.12	.24
Qwen3.5-27B	J – logit	0.045	[-0.195, 0.275]	0.776	.40	.19	.41
Gemma-3-27B-it	R – J	1.010	[0.785, 1.235]	$< 10^{-4}$	.76	.10	.14
Gemma-3-27B-it	R – logit	1.440	[1.170, 1.675]	$< 10^{-4}$	.80	.03	.17
Gemma-3-27B-it	J – logit	0.430	[0.055, 0.790]	0.112	.54	.06	.40

Table 1: Primary endpoint and the two supporting contrasts, under the mean-of-two-validated-judges rule. 100 cells and 20 prompts per model. Positive Δ favours the first arm.

On Gemma the bootstrap interval for J – logit excludes zero ([0.055, 0.790]) while the permutation test does not reject ($p = 0.112$). The two procedures ask slightly different questions and the permutation test is the pre-specified one, so we read that contrast as not significant — but the disagreement is reported rather than settled by taking whichever of the two is more favourable.

Absolute scores are low. Mean contextual coherence is 1.99 (Qwen) and 2.57 (Gemma) for R-Lens, against 1.30 and 1.56 for J-Lens and 1.25 and 1.13 for the logit lens (Figure 1a). On this rubric the mid-scale band is “partially relevant, not a coherent continuation”. R-Lens readouts at $z \leq 0.4$ are better than the alternatives and still not good; on Qwen the R-Lens mean does not reach the midpoint of the scale.

4.2 Depth

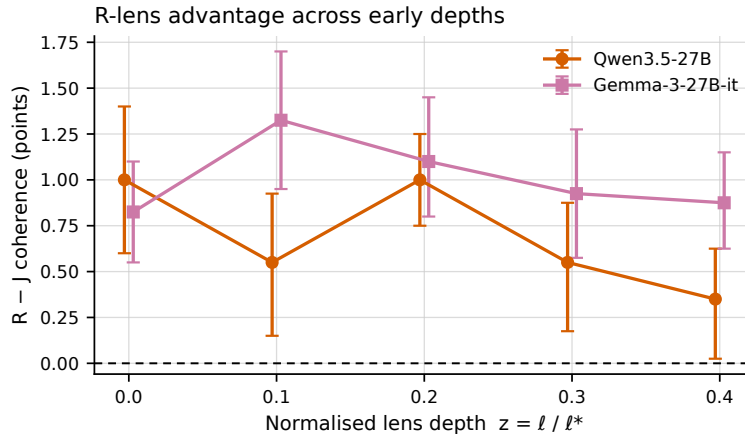


Figure 2: R – J by normalised depth, with 95% intervals. The two models do not agree on where in depth the advantage is largest.

The depth profiles differ between models. On Gemma the advantage holds at every depth and survives Holm correction at all five: +0.825 at $z = 0$, a peak of +1.325 at $z = 0.1$, +0.875 at $z = 0.4$. On Qwen it is largest early (+1.000 at both $z = 0$ and $z = 0.2$) and falls to +0.350 by $z = 0.4$; three of the five Qwen depths do not survive Holm correction ($z = 0.1, 0.3$ and 0.4 , all p_{Holm} between 0.06 and 0.08).

The effect therefore decays with depth on Qwen and is roughly flat on Gemma, and on neither model is it concentrated at the single earliest layer. What both models support is that R-Lens is

ahead across $z \leq 0.4$; a sharper claim about *where* within early depth the improvement sits is not supported by these two models together.

4.3 Secondary dimensions

Model	Dimension (R – J)	Δ	95% CI
Qwen3.5-27B	Lexical integrity	0.205	[0.125, 0.275]
Qwen3.5-27B	Prompt echo	0.245	[0.160, 0.320]
Gemma-3-27B-it	Lexical integrity	0.245	[0.150, 0.345]
Gemma-3-27B-it	Prompt echo	0.330	[0.240, 0.410]

Table 2: Secondary dimensions, both on 0–2 scales.

R-Lens produces more well-formed tokens (lexical integrity) *and* copies more of the prompt (prompt echo). Both differences exclude zero on both models. Section 5.4 asks whether the second explains the first.

We do not compare the magnitude of the echo difference to the magnitude of the coherence difference. They are measured on different scales (0–2 and 0–4) and such a comparison would be meaningless.

4.4 By evaluation set

Per-set differences are all positive on both models and range from +0.425 (Qwen typo; Gemma multilingual) to +2.150 (Gemma poetry). **No per-set difference survives Holm correction** ($p_{\text{Holm}} = 0.641$ throughout). With four prompts per set this analysis is descriptive only and is reported for pattern, not for inference. Poetry is the largest difference on both models, but on four prompts that is an observation, not a finding.

5 Robustness

5.1 Integrity audit

A fail-closed audit re-derived the frozen artifacts from scratch: it recomputed the panel hash, replayed the per-(cell, rater) arm permutation and compared it to the key, reconstructed all 400 outgoing judge payloads and scanned them for lens-identity leakage, and re-parsed every stored judge response.

Twenty conditions were checked. On the first pass nineteen held—including zero unparseable responses across 526 stored ratings, zero arm-mapping mismatches, and an exact reproduction of the panel hash (a4984e34d835c60ecb661552ec0a794e ...).

One condition failed, and it was ours. The score-combination rule derived the combined winner by *voting* on the judges’ winners while *averaging* their scores, which let the two disagree on 18 of 200 cells. The rule was corrected to derive the winner from the combined scores, and the frozen raw responses were re-combined without any new API call, so no cell was re-rated and no rating changed.

Re-running the full analysis on the corrected scores changed no reported number—all 18 cells were ties under both rules. The audit was then re-run against the corrected ratings and the analysis those ratings produced, and **all twenty conditions pass**, including

`winner_matches_max_score` across all 200 cells and `ratings_frozen_before_analysis`. A defect that changes no number is still reported here, because whether it changed a number is not something a reader should have to take on trust.

5.2 The adjudicator, and why it is not in the primary estimate

The original design adjudicated disagreements with a third judge. That judge was put through the same 78-cell battery as the primaries and cleared eight of nine gates outright: it never preferred a corrupted arm (0/10), showed no position bias ($\chi^2 = 0.66$ on 2 df), scored provably-identical arms identically, returned a tie on 5/5 identity-layer cells, showed full dynamic range on late-layer controls, and left 0/190 responses unparseable.

It did not clear the order-invariance gate. Admission used the frozen *operational* rule — an observed order-flip rate of at most 15% — not an inferential non-inferiority test; GPT-5 passed at 1/15 (6.7%) and the adjudicator failed at 10/53. Intervals are reported below because both estimates carry considerable uncertainty, and neither judge’s true rate is established. Rotating the arm order flipped the winner on **10 of 53 non-tied pairs (18.9%, 95% CI [9.4%, 32.0%])** against a 15% ceiling. An exact binomial test cannot reject 15% ($p = 0.266$), and the interval is far too wide to establish that the rate is acceptable. Escalating the battery does not help: resolving 18.9% against 15% at 90% power needs roughly 810 comparable pairs, a control panel several times the size of the experiment it exists to validate. For scale, the admitted primary judge GPT-5 scored 1/15 (6.7%) with an interval of [0.2%, 32%]—so this control cannot even separate the adjudicator from a judge already in use.

We therefore treated the unresolved gate as a failure and demoted the adjudicator, rather than admitting an instrument the battery could not clear or loosening the threshold to fit it. We do *not* claim to have shown the adjudicator is order-biased: the correct statement is that it did not satisfy a pre-specified validation gate, and the data cannot establish that its true rate is either acceptable or unacceptable. The primary estimate is the mean of the two validated judges. The adjudicated estimate is retained as a sensitivity analysis, and the adjudicator’s ratings are preserved rather than deleted, since they are evidence about the panel even when not used to score it.

Disclosure of timing. The battery and its 15% ceiling were fixed in the frozen protocol before any judge was run. The *consequence* of excluding the adjudicator, however, was already visible: the judge-dependence sensitivity analysis (§5.3), which reports the mean-of-two estimate at +0.850 against the adjudicated +0.778, was computed before the adjudicator’s battery was run. We therefore describe the demotion as a transparent post-validation analysis amendment against a pre-specified gate, not as a fully blind fail-closed decision. Both estimates appear throughout.

5.3 Judge dependence

A two-judge mean is still a scoring rule, and a different rule could give a different answer. Recomputing under four frozen rules—GPT-5 alone; DeepSeek alone; the mean of the two (primary); and the original adjudicated rule—gives a positive $R - J$ on every rule and on both models, with every individual-judge interval excluding zero. The pooled estimate ranges from +0.725 (DeepSeek alone) to +0.975 (GPT-5 alone), with the primary rule at +0.850 and the adjudicated rule at +0.778.

The spread across rules (0.25 points pooled) is smaller than the effect itself, and no rule places the interval anywhere near zero. **The adjudicated estimate—the one built on the instrument we demoted—is the smallest of the four**, so excluding the adjudicator did not manufacture the result; it removed a conservative influence on it.

R-lens advantage is present under every scoring rule

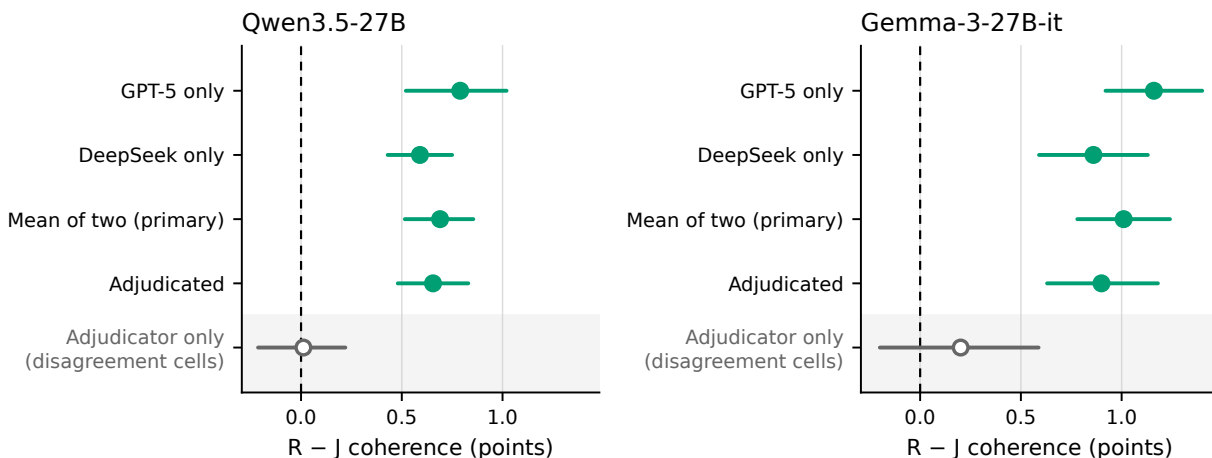


Figure 3: R - J under four scoring rules. “Mean of two” is the primary rule; “Adjudicated” is the original rule, retained as a sensitivity analysis after its third judge failed to clear validation. The shaded band is a diagnostic computed only on cells where the two primary judges disagreed; it is conditioned on disagreement and is not a fifth estimate of the same quantity.

The logit-lens contrasts are not stable. That verdict is scoped to R - J, and the scoping is load-bearing. Checking every contrast against the same two single-judge intervals, **five of nine are not stable across the two primary judges**. On Qwen, GPT-5 rates J-Lens *significantly worse* than the logit lens (-0.680 , $[-1.06, -0.33]$) while DeepSeek rates it *significantly better* ($+0.770$, $[0.57, 0.97]$)—same cells, disjoint intervals, opposite signs. Every unstable contrast involves the logit lens; every R - J contrast is stable.

One plausible explanation is a floor effect: the judges may agree about the relative ordering of two Jacobian transports while anchoring the bottom of a 0–4 scale differently when scoring near-noise. We have not tested that mechanism, and the pattern is also consistent with rubric underdetermination at the low end. Any comparison against the logit lens in this report should be read with both judges’ values in view and must not be quoted as a single number.

Agreement. Quadratic-weighted $\kappa = 0.514$ on 600 paired scores, exact agreement 39.3%, mean absolute difference 0.91 points. The two validated judges disagreed on 63% of cells—the disagreements the third judge was introduced to settle, and which the primary rule now simply averages over. **This is moderate agreement at best.** Validation reduces the concern that this is gross judge failure, but it does not establish that the disagreement originates in the readouts. It may reflect genuine ambiguity in early-layer readouts, differences in judge calibration, or residual underdetermination in the rubric. A κ of 0.51 is not a precise instrument, and every interval in this document inherits that noise.

5.4 Prompt echo

R-Lens copies more of the prompt than J-Lens, and a rater who rewards prompt-shaped text would score it higher for that reason alone. Restricting to cells where both lenses received the *same* echo score:

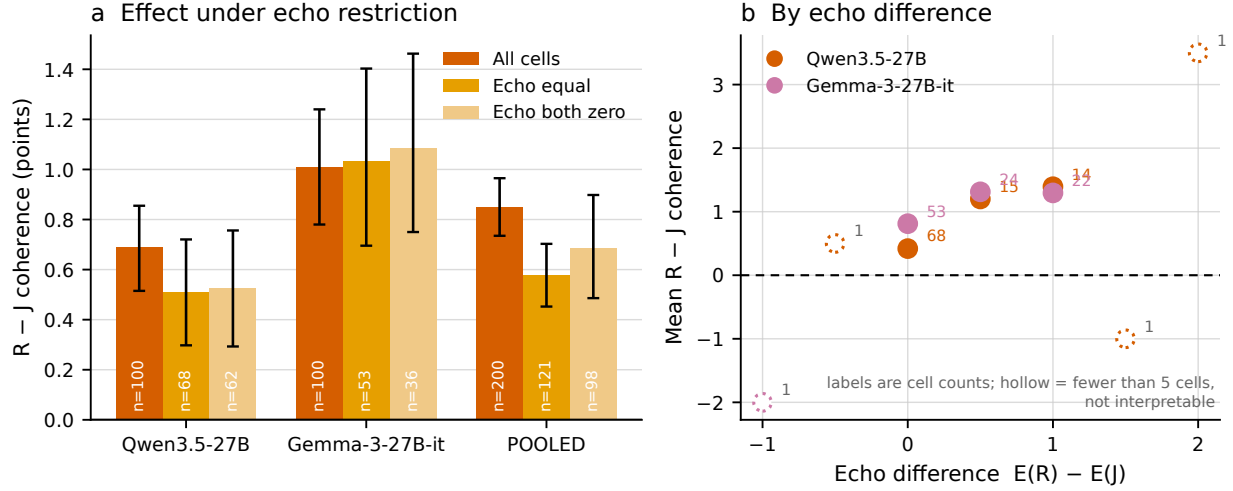


Figure 4: Echo sensitivity. (a) $R - J$ restricted by echo agreement, retained cell counts printed inside each bar. (b) Mean $R - J$ by echo difference; labels are cell counts and hollow markers mark strata too small to interpret.

Model	All cells		Echo equal retained		Echo both zero retained	
Qwen3.5-27B	0.690	0.509 [0.30, 0.72] (n=68)	74%	0.526 [0.29, 0.76] (n=62)	76%	
Gemma-3-27B-it	1.010	1.031 [0.70, 1.40] (n=53)	102%	1.083 [0.75, 1.46] (n=36)	107%	
Pooled	0.850	0.577 [0.45, 0.70] (n=121)	68%	0.686 [0.49, 0.90] (n=98)	81%	

Table 3: $R - J$ contextual coherence under echo restriction, frozen mean-of-two scoring. “Retained” is the restricted estimate as a percentage of the all-cells estimate; it can exceed 100%, as it does on Gemma, where matching on echo does not shrink the effect at all.

The effect survives in all 24 subset estimates—every scoring variant, both models, both restriction rules—with intervals excluding zero throughout. **But roughly a third of the pooled gap coincides with an echo difference** under the equal-echo restriction, and the pattern is not uniform: on Qwen the estimate falls by about a quarter, while on Gemma it does not fall at all. The Gemma both-zero cell retains only 36 cells from 17 prompts and its interval is correspondingly wide; it should not be read as evidence that echo is irrelevant there.

This is not a causal adjustment. Echo is measured from the same readouts as coherence, so restricting on it conditions on a post-treatment variable. The regression of the coherence difference on the echo difference is descriptive; its intercept is the fitted difference at equal echo, not an echo-adjusted effect. **The two judges also disagree about the mechanism:** GPT-5’s slope is ≈ 1.0 – 1.1 with intervals excluding zero on both models, while DeepSeek’s is 0.29 $[-0.16, 0.80]$ and 0.35 $[-0.03, 0.63]$, both covering zero. GPT-5 sees echo and coherence tightly coupled; DeepSeek sees almost no coupling. Both still rate R above J .

Read this analysis as a caution, not as reassurance. Restricting to cells with equal observed echo selects on a variable that lens identity itself affects. Conditioning on a post-treatment variable can distort a comparison and does not estimate the effect with echo removed. The following section measures the quantity directly, and gives a substantially less reassuring answer than this restriction does — which is the point of reporting both.

5.5 An exploratory attempt to exclude prompt-local content, and why it did not validate

Read this section as a measurement study, not a second endpoint. It was motivated after seeing the prompt-echo result, its rubric was frozen before any rating, and it then failed a validation control. We report it in full because the failure is informative, but no claim in §11 rests on its estimates.

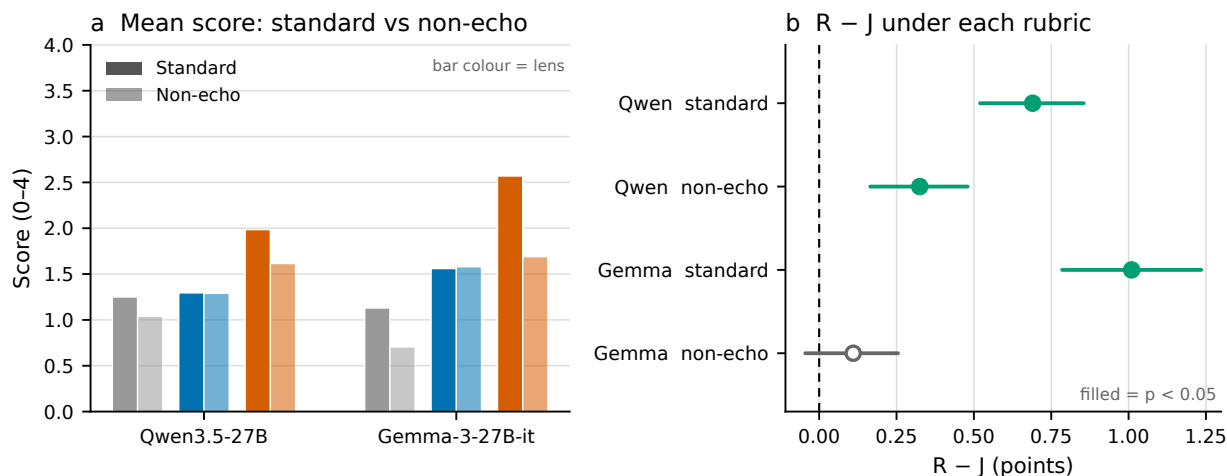


Figure 5: Standard versus non-echo scoring. (a) Mean score by lens under each rubric, same 0–4 scale. (b) $R - J$ under each rubric; a filled marker denotes $p < 0.05$ on the pre-specified permutation test. The R-Lens bar falls sharply under non-echo scoring on both models while the J-Lens bar barely moves.

A second rubric was written, hashed and frozen under its own salt. It instructs the judge to discard every token that repeats a word visible in the prompt, including trivial variants, and to score only the remainder on the same 0–4 scale; a readout that is entirely repetition has nothing left and scores 0.

What the rubric was checked against before use. Ten control cells pitted a pure prompt-copy arm — the prompt’s own final tokens — against real late-layer readouts. Both judges scored the copied arm **0.00 out of 4** and it won **0 of 10** cells. Rating was blocked in software until that passed, and the control is keyed to the rubric’s hash, so editing the rubric invalidates its validation. All 200 panel cells were then rated by both judges with zero parse failures.

That control was too easy, and we did not realise it at the time. Rejecting a readout that is *entirely* a prompt copy is a much simpler task than scoring one that *mixes* copied and novel tokens — which is what every real readout in the panel is. Section 5.7 reports the control that tests the harder case, run afterwards, which one of the two judges failed.

The attenuation percentages are not stable quantities. We quote 53% (Qwen) and 89% (Gemma) because they are computed from the pooled primary estimates, but they are ratios of two noisy numbers and on Gemma the denominator and numerator disagree between judges. Recomputing each judge’s attenuation from that judge’s own two estimates:

Model	GPT-5 alone	DeepSeek alone	Pooled (quoted)
Qwen3.5-27B	0.790 → 0.490 = 38%	0.590 → 0.160 = 73%	53%
Gemma-3-27B-it	1.160 → 0.500 = 57%	0.860 → -0.280 = 133%	89%

Model	Contrast	Δ	95% CI	p
Qwen3.5-27B	R – J	0.325	[0.165, 0.480]	0.0042
Qwen3.5-27B	R – logit	0.575	[0.415, 0.715]	$< 10^{-4}$
Qwen3.5-27B	J – logit	0.250	[0.025, 0.465]	0.070
Gemma-3-27B-it	R – J	0.110	[-0.045, 0.255]	0.450
Gemma-3-27B-it	R – logit	0.985	[0.805, 1.180]	$< 10^{-4}$
Gemma-3-27B-it	J – logit	0.875	[0.685, 1.065]	$< 10^{-4}$

Table 4: Non-echo coherence. Same estimand, estimator, bootstrap and permutation test as Table 1; a different scoring construct.

Model	Non-echo coherence (0–4)			Residual substance (0–2)		
	Logit	J-Lens	R-Lens	Logit	J-Lens	R-Lens
Qwen3.5-27B	1.04	1.29	1.61	1.57	1.93	1.78
Gemma-3-27B-it	0.71	1.58	1.69	1.81	1.81	1.48

Table 5: Mean non-echo coherence and mean residual substance by lens. Residual substance is the judge’s *report* of how much non-copied material there was to score, not a quality judgement and not a direct count. R-Lens is reported lower than J-Lens on both models; on Qwen the logit lens is lower still, and on Gemma R-Lens is the lowest of the three. This is why the attenuation cannot be read as loss of novel-content quality alone.

DeepSeek’s Gemma attenuation exceeds 100% because its non-echo estimate is negative: the “attenuation” there is a sign reversal, not a shrinkage. **The 89% figure should therefore be read as “somewhere between 57% and a reversal, depending on which judge you ask”, not as an estimate.** We retain the pooled figures for comparability with the primary rule and state their per-judge range wherever they appear. The Qwen range (38–73%) is wide but at least sign-consistent.

On Gemma the R-Lens advantage over J-Lens does not survive. The estimate falls from +1.010 to +0.110 and the interval includes zero. The pooled cell-level win/tie/loss is nearly balanced at 46/27/27 — but the per-judge analysis below shows this masks a sign reversal between judges, not shared evidence of equivalence. On Qwen an advantage survives at less than half its size, +0.690 $\rightarrow$ +0.325, still excluding zero.

We state these as *attenuations of the measured gap*, not as causal decompositions: the R – J gap attenuates by 53% on Qwen and 89% on Gemma under a rubric that excludes prompt-overlapping content. It does not follow that 53% or 89% of the original effect *is caused by* echo. The two rubrics measure different constructs, and the non-echo rubric additionally penalises a lens whose copied tokens occupy slots in a fixed-length top-10 list — an effect we could separate only by refilling the list from deeper rankings, which we did not do (§11).

Judge robustness of the non-echo result. The standard result is reported under four scoring rules (§5.3); the non-echo result needs the same treatment, since the two judges already reverse in sign on J – logit under the standard rubric. Recomputing R – J from each judge’s non-echo ratings alone:

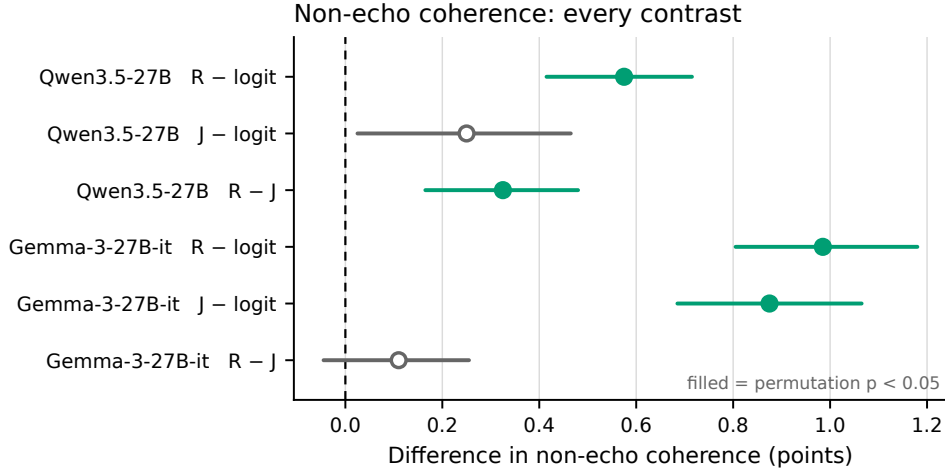


Figure 6: Every non-echo contrast. Both Jacobian lenses beat the logit lens on Gemma and R-Lens does on Qwen, while R - J is inconclusive on Gemma. A filled marker denotes significance on the pre-specified permutation test, which is why the Qwen J - logit row is hollow despite its interval excluding zero ($p = 0.070$).

Model	GPT-5 alone	DeepSeek alone	Mean of two (primary)
Qwen3.5-27B	+0.490 [0.21, 0.76] $p = 0.016$	+0.160 [0.03, 0.29] $p = 0.075$	+0.325 [0.17, 0.48] $p = 0.004$
Gemma-3-27B-it	+0.500 [0.30, 0.69] $p = 0.006$	-0.280 [-0.44, -0.12] $p = 0.085$	+0.110 [-0.05, 0.26] $p = 0.450$

On Gemma the two judges disagree in sign, with disjoint intervals. GPT-5 puts R-Lens half a point ahead; DeepSeek puts it more than a quarter of a point behind. The pooled +0.110 is the average of a disagreement, not a shared finding of no effect — and those are very different things. On Qwen both judges are positive: GPT-5’s estimate reaches $p = 0.016$ while DeepSeek’s does not reach the pre-specified threshold ($p = 0.075$).

The non-echo instrument is markedly less reliable than the standard one. Quadratic-weighted $\kappa = 0.19$ against 0.51 for the standard rubric, exact agreement 25.8% against 39.3%, mean absolute difference 1.44 points against 0.91 on the same 0–4 scale. A κ of 0.19 is slight agreement. Every non-echo number in this section inherits that, and we did not anticipate it: the copy control established that judges reject a pure copy, which turns out to be a much easier task than scoring what remains after partial copying.

The non-echo intervals understate their own uncertainty. The bootstrap resamples *prompts*, which is correct for the estimand, but the score entering each cell is the mean of two judges whose agreement is slight. The intervals therefore propagate prompt-level variation and *not* judge disagreement. The per-judge rows above are the honest expression of that second source, and they lie much further apart than any single interval implies. We also computed no agreement statistic for *residual substance* itself; the κ above is for the primary non-echo dimension only.

Judges report less residual material for R-Lens. The non-echo rubric also asks each judge how much non-copied material was there to score (*residual substance*, 0–2). R-Lens is rated *lower* than J-Lens on both models: $-0.330 [-0.395, -0.265]$ on Gemma and $-0.150 [-0.195, -0.105]$ on Qwen, both excluding zero.

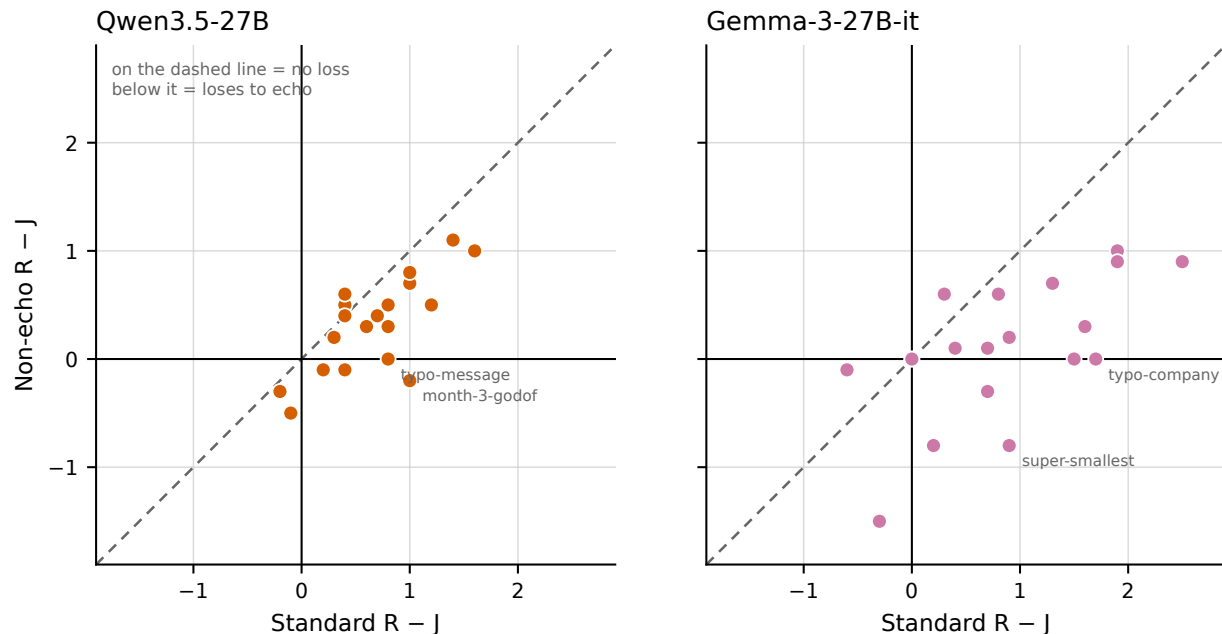


Figure 7: Attenuation is systematic, not driven by a few prompts. Each point is one prompt’s $R - J$ averaged over its five depths, standard scoring on the x axis against non-echo scoring on the y . A prompt on the dashed $y=x$ line loses nothing when copied tokens are excluded. On Qwen most prompts sit modestly below the line; on Gemma they sit much further below it and more variably, with several large standard effects collapsing to zero or negative.

This is *judge-reported* evidence consistent with the crowd-out concern in §11: the non-echo comparison varies both the apparent quality and the apparent quantity of remaining content. The attenuation therefore conflates the coherence of the remaining content with the amount of content the judges perceive as remaining, and the present design cannot apportion the two. Doing so would require refilling each list from deeper rankings, which we did not do.

We report residual substance as a rating, not as a measurement. With $\kappa = 0.19$ on the primary non-echo dimension, and no agreement statistic computed for residual substance itself, it should not be objectified as a count of tokens.

What strengthens under non-echo scoring. R-Lens beats the logit lens on both models; J-Lens clearly does so on Gemma ($+0.875$, $p < 10^{-4}$, against $+0.430$ under standard scoring) and suggestively on Qwen ($+0.250$, $[0.025, 0.465]$, $p = 0.070$ — the interval excludes zero but the pre-specified permutation test does not reject). J-Lens echoes less than R-Lens, so standard scoring was penalising it for content the non-echo rubric ignores.

We stop short of concluding that the strongest evidence is about Jacobian transport in general. Across both rubrics: $R - \text{logit}$ is strong on both models; $J - \text{logit}$ is strong on Gemma under non-echo scoring, inconclusive on Qwen under non-echo scoring ($p = 0.070$), and non-significant on both models under standard scoring, where it also reverses in sign between judges. **The data do not support a uniform cross-model attribution to the Jacobian transport alone.** What they support is narrower: on Gemma both Jacobian methods beat the logit lens on non-copied content, and on Qwen only R-Lens clearly does.

Restriction versus rubric-based exclusion. These are two different repairs for the same confound, and neither works cleanly. Restriction (§5.4) retained 74% of the effect on Qwen and 102% on Gemma; rubric-based exclusion retains 47% and 11%. Conditioning on equal observed echo and excluding echoed content answer different questions: the former selects on a lens-affected variable, the latter changes the scoring construct. Neither identifies a formal mediation fraction.

Restriction conditions on a post-treatment variable and so cannot estimate a direct effect. Rubric-based exclusion avoids that specific error but introduces two of its own: rater instability (§5.7) and confounding of content quality with content quantity. **Neither method cleanly identifies the contribution of prompt-local content**, and the sharper methodological result is that both plausible repairs fail in different ways — which is more useful to a reader than a claim that one of them worked.

5.6 Judge behaviour

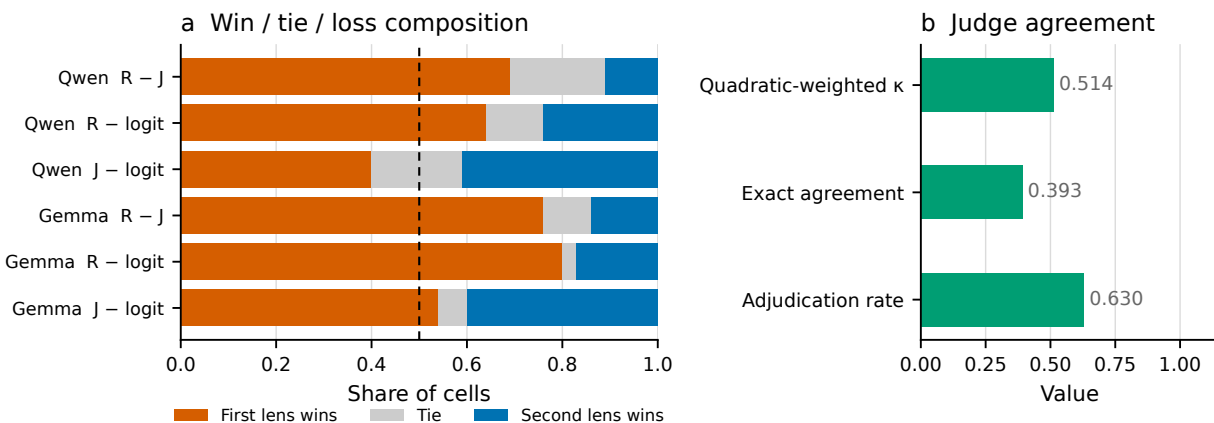


Figure 8: What the judges did. (a) Win/tie/loss composition per contrast under the primary rule. (b) Agreement statistics. A pooled mean difference of 0.85 points is compatible with very different underlying distributions; the composition shows which.

R-Lens wins 69% (Qwen) and 76% (Gemma) of R – J cells and loses 11% and 14% respectively. The effect is a broad shift across the panel, not a handful of extreme cells — so it is not the product of a few outlier readouts that a different rubric might score differently.

5.7 Mixture invariance: the control the rubric failed

The rubric instructs judges to *discard* prompt-local tokens. If they do, adding more of them to a readout whose meaningful content is unchanged must not move the score. That is a testable property, and it is the property the panel actually depends on.

We built 20 families. Each holds a **fixed ten-token coherent core** — drawn only from arms that had already scored ≥ 2 on this rubric, so every family has something to lose — and varies only the number of prompt-local distractors mixed in: 0, 3 or 6, at **constant list length ten**. Varying length would confound length with padding. Three padding kinds are tested separately: literal repetition, case variants, and near-miss spellings of prompt words. Acceptance criteria were frozen in code before any rating: median within-family spread ≤ 1.0 on the 0–4 scale, and at most 34% of families declining monotonically as padding increases.

Judge	Median within-family spread	Monotone decline	Verdict
GPT-5	1.00 (max 1.00)	2/20 = 10% (max 34%)	PASS
DeepSeek-v3.1	2.00	10/20 = 50%	FAIL

One judge follows the instruction and the other does not. DeepSeek’s scores fall monotonically with padding in half of all families, on cores it had itself rated as coherent. Its per-padding-kind spreads are 1.00 for literal repetition, 1.00 for case variants and 1.67 for near-miss spellings — so the effect is largest exactly where “prompt-local” is most a matter of judgement. GPT-5 passes, but sits precisely on the ceiling at 1.00.

We do not drop DeepSeek and keep GPT-5. Selecting the judge that passes, after seeing which one passes, is outcome-dependent instrument selection, and it would be the most attackable decision in this document. Both judges were admitted on the same pre-specified battery and both are retained.

An earlier version of this control was weaker than it looked. Our first attempt did not require the core to be coherent, and drew it from whichever arm the panel happened to list first — newline tokens, for poetry cells. Both judges “passed” it, but DeepSeek scored 11 of 20 families at zero throughout, where the spread is zero however the judge behaves. Restricting that run to the three families where anything scored above 1 already showed a median spread of 2.00 and 2/3 declining monotonically. The corrected control above reproduces that at $n = 20$. We report the first attempt because its failure mode — a control that passes by measuring nothing — is worth recording.

What this costs the section. With $\kappa = 0.19$, a sign disagreement on Gemma and one judge failing mixture invariance, the non-echo estimates cannot support a claim about how much of R-Lens’s advantage survives the exclusion of prompt-local content. **What they support is the negative result: explicitly excluding prompt-local content did not produce a measure two validated judges agree on.** The decomposition is measurement-sensitive, and that is worth knowing before anyone builds on it.

6 What the readouts actually look like

Scores alone ask a reader to take the instrument on trust. Three cells are shown below, chosen by *post-hoc illustrative* rules applied to the frozen panel rather than by eye — illustrative, not confirmatory: the largest combined standard-and-non-echo R-Lens gain, the largest gap between a standard gain and its non-echo counterpart, and the cell where all three arms score lowest. Tokens are shown exactly as the judges saw them.

A genuine gain. Qwen, association, $z = 0$. Prompt: “*They shook hands for the cameras and held it a beat too long, each waiting for the other to let go first.*”, read at the final period.

Lens	Top-10 readout	Contextual	Echo	Non-echo
Logit	· , (- / : ?	1.0	0.0	1.5
J-Lens	--and -- • **-- tossed .hm izon	0.0	0.0	1.0
R-Lens	... though ... they seperate there wanting needing	3.0	0.0	3.5

R-Lens surfaces *wanting*, *needing* and *though* — a coherent reading of a standoff — alongside the prompt-overlapping pronoun *they*, which does appear in the prompt (“*They* shook hands”). The judges nonetheless assigned this arm an echo score of 0.0, defensible for a single function word but not literally zero overlap; we report it as scored rather than adjust it. Its non-echo score (3.5) is *higher* than its standard score (3.0). This is what the claim looks like when it holds.

An echo-driven gain, where the two rubrics disagree about the winner. Gemma, typo, $z = 0.2$. Prompt: “*The stadium was packed with nearly forty thousand poeple*” — a misspelling — read at eple.

Lens	Top-10 readout	Contextual	Echo	Non-echo
Logit	ized ed an al Than - fromi Tourist Min	0.5	0.0	1.0
J-Lens	recieve seperate succesfully wouldnt couldnt havent	0.5	0.0	2.0
R-Lens	, and people in ' seperate (to " '	3.0	0.5	0.5

The two rubrics rank these arms in opposite orders. R-Lens recovers *people* — the intended word, which is prompt-local — and the standard rubric scores it 3.0 against J-Lens’s 0.5. J-Lens instead surfaces a cluster of *other common misspellings* (*recieve*, *seperate*, *succesfully*, *wouldnt*), which is not prompt-local at all, and the non-echo rubric scores it 2.0 against R-Lens’s 0.5.

This is why “prompt echo” is too narrow a name. *people* is not a copy: the prompt contains *poeple*. R-Lens is performing *orthographic normalisation* — reconstructing the intended word from a corrupted one — which is plausibly real computation, and on the **typo** set is arguably the task itself. Our rubric nonetheless discounts it, because it asks the judge to discard anything recoverable from the prompt. The standard rubric conflates four distinguishable things:

Construct	Example from this cell
Literal echo	repeating <i>stadium</i> , <i>thousand</i>
Prompt-local normalisation	<i>poeple</i> → <i>people</i>
Abstract contextual representation	other common misspellings
Novel downstream content	anything not locally recoverable

Our non-echo rubric removes credit for the first two together. That is defensible when the question is “what is here beyond prompt-local content”, but it is not the same as measuring copying, and the attenuations in §5.5 should be read as loss of *prompt-local content*, literal and normalised together. The source post explicitly treats readouts representing the current or similar tokens as evidence of early-layer structure rather than as a confound, so on its own framing R-Lens’s behaviour here is a success rather than an artefact.

This cell illustrates the central construct distinction: a one-dimensional coherence score silently chooses between prompt-local reconstruction and abstract contextual representation, and that choice determines which lens wins.

Where nothing works. Qwen, poetry, $z = 0.1$, read at the newline ending line one. Every arm scores 0.0 on contextual coherence; J-Lens and R-Lens both return multilingual fragments (*zeitig jenigen, beau alak edal ISTR*). Roughly a fifth of R – J cells are ties, and this is what they look like.

One anomaly worth recording. In this cell the logit lens scored 0.0 on contextual coherence but 2.0 on non-echo coherence, for a readout consisting entirely of punctuation and whitespace. That

ordering is hard to defend and is an instance of judge noise on degenerate inputs. This is *cross-rubric* instability on a degenerate input, distinct from the between-judge disagreement summarised by $\kappa = 0.51$, which measures agreement between two raters on one rubric. In aggregate the logit lens’s mean non-echo score is *below* its mean standard score on both models (1.04 against 1.25 on Qwen, 0.71 against 1.13 on Gemma), so this is not the general pattern — but we observed one such case and did not estimate its frequency among punctuation-dominated readouts.

7 Small-sample stability

The inferential unit is the prompt, not the cell: 20 prompts per model, each contributing five depths and three arms. One hundred cells can conceal a result that rests on one unusual prompt, so each prompt — and then each evaluation set — was deleted in turn and the estimate recomputed. Deleting a prompt deletes all of its cells, because the prompt is the bootstrap cluster.

Construct	Model	Leave-one-prompt-out	Leave-one-set-out	Prompt sign test
Standard	Qwen	0.600–0.700 (20/20 >0)	0.556–0.706 (5/5)	18+/2−, $p = 0.0004$
Standard	Gemma	0.820–0.973 (20/20 >0)	0.650–1.006 (5/5)	16+/2−, $p = 0.0013$
Non-echo	Qwen	0.273–0.380 (20/20 >0)	0.263–0.394 (5/5)	14+/5−, $p = 0.064$
Non-echo	Gemma	0.052–0.172 (20/20 >0)	−0.081–0.281 (4/5)	11+/5−, $p = 0.210$

Table 6: $R - J$ under exhaustive single deletions. The sign test is a two-sided exact binomial over per-prompt effects, each averaged across that prompt’s five depths; it is descriptive and discards magnitude, so it is less powerful than the primary estimator and is reported beside it, not instead.

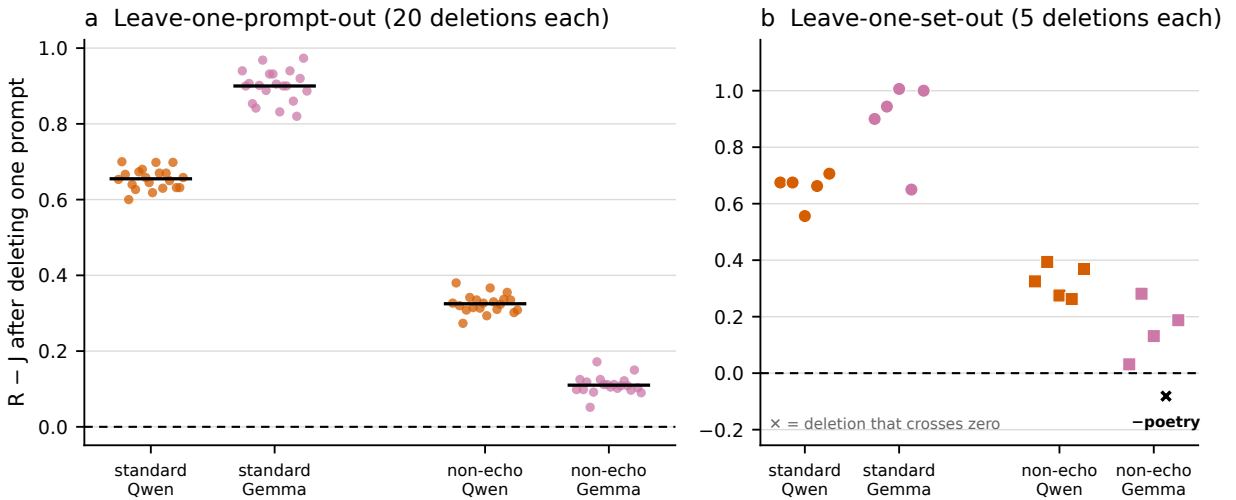


Figure 9: Exhaustive single deletions. (a) Every leave-one-prompt-out estimate; the bar is the mean of the twenty. (b) Every leave-one-set-out estimate. The one deletion that crosses zero — Gemma, non-echo, poetry removed — is marked. Panel (a) shows the standard estimates sitting far from zero on both models while the non-echo Gemma cloud sits just above it.

The standard result is not driven by any one prompt or set. On both models $R - J$ stays positive after deleting any single prompt and after deleting any single evaluation set, and the

prompt-level sign test is significant on both. The $n=20$ objection does not undermine the standard finding.

The non-echo results are not equally stable, in two specific ways. First, **poetry carries a disproportionate share of the Gemma effect**. Removing it drops the standard Gemma estimate to $+0.650$, the lowest of the five deletions, and takes the non-echo Gemma estimate *negative* (-0.081). Poetry is also the set with the largest per-set difference on both models (Table in §4.4) and the only set read at a different position — the newline ending line one, per the source protocol, rather than the final prompt token. We do not have the prompts to determine whether that position, or the genre, is responsible.

Second, **the prompt-level sign test does not confirm the Qwen non-echo result** ($p = 0.064$, 14 of 19 non-tied prompts positive), even though the primary estimate excludes zero ($+0.325$, $[0.165, 0.480]$, $p = 0.0042$). The two are not in contradiction — the sign test discards magnitude and is the weaker instrument — but the Qwen non-echo advantage rests more on the size of the differences than on a clear majority of prompts, and should be described that way.

Taken together: the standard-rubric ordering is robust, and the pooled non-echo estimates are positive but fragile. On Gemma that positive pooled estimate masks opposite-signed judge estimates; on Qwen both judges are positive but only GPT-5 reaches $p < 0.05$. The non-echo findings are therefore not directionally consistent across judges, and the fragility is compounded rather than merely observed.

8 The earlier no-prompt panel

An earlier 50-cell panel on Qwen returned a null for $R - J$ ($+0.080$, 95% CI $[-0.16, 0.34]$, $p = 0.596$). It is reported here for completeness and is **not comparable** to the present result. The two panels differ simultaneously in prompt visibility (that panel did not show raters the prompt), size (50 vs 200 cells), rubric scale, judge count, sampling rule and adjudication, and one of its raters was contaminated. No single one of those differences can be identified as the reason the results differ, and we do not claim to know which mattered. It is a preliminary attempt that was superseded, not a contradicting measurement.

That panel did produce one durable negative result, obtained against 150 human judgements: of three cheap token-form proxies for coherence, only the “trash-rate” proxy correlated in the expected direction ($\rho = -0.640$). Zero-corpus-frequency rate ($\rho = +0.330$) and in-prompt rate ($\rho = -0.245$) were both *inverted*, retracting two earlier readings of ours. Token-form statistics are not a substitute for rating the readouts.

9 Limitations

1. **One judge was demoted, not merely unused.** The adjudicator cleared 8 of 9 validation gates but its order-invariance rate (18.9%, $[9.4\%, 32.0\%]$) could neither be admitted nor rejected by a feasible battery. We excluded it. A reader who would have admitted it should read the adjudicated row of Figure 3 as the headline instead; it is smaller ($+0.778$ pooled) and still excludes zero.
2. **Order invariance is weakly controlled for every judge.** The same battery gives GPT-5 an interval of $[0.2\%, 32\%]$. The control is too blunt at this size to certify any of the three judges tightly, and a larger one was judged not worth its cost.

3. **Released lenses use the 25-prompt light recipe**, not the 1000-prompt recipe of the source post. Effect sizes are not comparable to the authors’. R-Lens and J-Lens may also differ in sample efficiency, so matching both arms at $n=25$ makes the comparison internally fair without making it equivalent to the full-recipe comparison.
4. **Model choice was fixed by the capstone brief**: Qwen3.5-27B and Gemma-3-27B-it. We did not verify that these are the exact revisions used in the source post, and make no claim that they are. All results are reported for these artifacts.
5. **Twenty prompts per model**. Adequate for the pooled estimand, which is why intervals are prompt-clustered; not adequate for the per-set breakdown, which is descriptive only.
6. **Autoraters, not humans**. No human ratings were used in the primary experiment. A superseded exploratory panel collected 150 human judgements and is reported in §8 only as historical context for a negative result about token-form proxies; none of it enters any estimate here. Judges were validated against controls, but LLM raters may share biases with each other and with the models being read, and no external human anchor establishes that their “contextual coherence” matches the construct a human interpretability researcher cares about.
7. **Echo is not causally controlled**, only restricted on.
8. **Logit-lens comparisons are judge-dependent** and should not be quoted as single numbers.
9. **Depth-localisation does not replicate across models**.
10. **Ordering, not quality**. Nothing here shows early-layer R-Lens readouts are coherent in absolute terms; they average roughly 2 of 4, and under non-echo scoring less than that.
11. **The non-echo rubric penalises crowd-out as well as copying**. Copied tokens both earn no credit *and* occupy slots in a fixed-length top-10 list that novel tokens could have filled. Separating those would require refilling each list from deeper rankings; the frozen panel stored only the top 10, so we did not. Crowd-out may cause this analysis to overstate the loss of novel content, but adding lower-ranked tokens can move an autorater’s score non-monotonically, so the attenuation is not a formal bound.
12. **Four prompts per evaluation set**. Per-set results are descriptive and none survives Holm correction.
13. **One rubric, one panel per construct**. We did not test whether a differently-worded non-echo rubric gives the same attenuation.
14. **Poetry is influential on Gemma**. It is the largest per-set difference on both models and the only set read at a different position. Removing it takes the Gemma non-echo estimate negative. Four prompts is too few to say whether the position, the genre, or chance is responsible.

10 Conclusion

We evaluate R-Lens’s early-layer contextual-coherence claim on Qwen3.5-27B and Gemma-3-27B-it using a blinded, pre-specified and frozen 200-cell panel scored by two independently validated LLM autoraters. Under a standard prompt-aware rubric the claimed advantage reproduces on both models and is robust: under that rubric the R – J direction survives every judge-aggregation rule we examined, the deletion of any single prompt, and the deletion of any single evaluation set.

R-Lens, however, also surfaces significantly more prompt-overlapping content. Under a narrower rubric that discards prompt-local tokens, the R – J advantage falls by roughly half on Qwen and becomes inconclusive on Gemma.

That second instrument did not validate. Its inter-judge agreement is slight ($\kappa = 0.19$ against 0.51); on Gemma its two judges disagree in *sign* with disjoint intervals; one of the two failed a

mixture-invariance control, its scores declining monotonically as ignorable padding was added to a fixed core; and R-Lens is rated as leaving less material to score in the first place. We therefore report it as an exploratory measurement study whose result is negative: **explicitly excluding prompt-local content did not produce a measure two validated judges agree on.**

The conclusion is neither a clean confirmation nor a refutation. R-Lens’s advantage under ordinary coherence scoring is real and reproducible, but ordinary coherence conflates novel content with *prompt-local* content — literal repetition and normalised reconstruction alike, and the latter may be genuine computation rather than copying.

We read this as a finding about the construct rather than about the method. A one-dimensional coherence score silently chooses between prompt-local reconstruction and abstract contextual representation, and §6 shows a single cell where that choice reverses which lens wins. Neither choice is obviously wrong; the source post treats current-token representation as evidence of early-layer structure, and on that framing R-Lens’s behaviour is a success. What we can say is that “coherence”, measured one-dimensionally, does not separate these, and that the ranking of these lenses is not invariant to which one you mean.

11 Supported and unsupported claims

Supported by these data.

1. Under the standard contextual rubric, R-Lens receives higher coherence scores than J-Lens on both models, and the direction holds for each validated judge separately.
2. R-Lens surfaces significantly more prompt-overlapping content than J-Lens on both models.
3. The R – J gap is substantially smaller under a rubric that excludes prompt-overlapping content.
4. Excluding prompt-local content did not produce a measure two validated judges agree on: $\kappa = 0.19$, a sign disagreement on Gemma, and one judge failing mixture invariance. This is reported as a negative result about the construct, and no lens comparison rests on it.
5. On non-copied content, R-Lens outperforms the logit lens on both models, and J-Lens does so clearly on Gemma with positive but inconclusive evidence on Qwen ($p = 0.070$). R – logit is consistently positive on both models; J – logit is strong only on Gemma under non-echo scoring and is inconclusive or judge-dependent everywhere else.
6. The standard-rubric R – J difference survives deletion of any single prompt and any single evaluation set, on both models (§7).

NOT supported, and not claimed.

1. That R-Lens is universally more semantically informative than J-Lens.
2. That a specific percentage of the standard R – J effect *is caused by* prompt echo. The attenuations are construct changes, not mediation fractions.
3. That restricting to equal observed echo identifies the direct effect of lens type. It conditions on a lens-affected variable.
4. That the source post’s $n=1000$ fitting recipe was replicated. We evaluate the released $n=25$ light-recipe artifacts.
5. That autorater scores equal human judgement, or that the rubric is a validated measure of interpretation quality. The control battery shows the judges follow the intended behaviours; it does not validate the construct.
6. That J-Lens consistently beats the logit lens under the standard rubric. That contrast is judge-dependent and reverses in sign between judges.

7. That the excluded adjudicator was shown to be order-biased.
8. That a non-significant result establishes equality.
9. That five depths per prompt constitute independent observations. The inferential unit is the prompt: 20 per model.
10. That the non-echo findings are as stable as the standard ones. The non-echo rubric has $\kappa = 0.19$; on Gemma the two judges disagree in sign; the Gemma estimate turns negative when the poetry set is removed; and neither model’s non-echo effect is confirmed by a prompt-level sign test at $\alpha = 0.05$.
11. That the Gemma non-echo null demonstrates the absence of an advantage. It reflects disagreement between two judges, not agreement that there is nothing there.
12. That the non-echo attenuation measures loss of novel content alone. R-Lens also has less residual material to be scored, and the two are not separable without refilling from deeper rankings.

12 What would change the conclusion

- A human panel of even modest size disagreeing with the autoraters on the standard rubric would undercut every coherence claim here at once, since nothing in this document is anchored to human judgement.
- A refilled non-echo analysis — filtering prompt copies from a deeper ranking and topping the list back up to ten — would separate loss of novel content from top-10 crowd-out. If the Gemma null persisted after refilling, the reading would firm up to “R-Lens has no detectable advantage in novel content on Gemma”. If it became positive, the attenuation would be largely a presentation effect.
- A mixture-invariance control — holding the non-copied tokens fixed while varying the number of copied tokens at constant list length — failing would show the non-echo rubric is presentation-sensitive, and its attenuations could not be read as clean exclusions of echo.
- Refitting both lenses at $n = 1000$ and finding the gap disappears would indicate the effect is a property of the light recipe.

13 Reproducibility

Every number in this document is produced by a command in the repository operating on frozen artifacts; none is transcribed by hand into a plotting script or a table. The figures are drawn by **rlens figures** directly from the analysis outputs, and skip themselves by name if an input is missing rather than falling back to defaults.

Stage	Command
Preflight (fail-closed)	rlens preflight
Prompt selection	rlens eligibility
Panel freeze	rlens freeze-panel, rlens panel-v2
Judge validation	rlens judge-validate
Rating	rlens autorate
Primary analysis	rlens analyse-v2
Integrity audit	rlens audit-v2
Judge and echo sensitivity	rlens robustness
Non-echo rubric validation	rlens non-echo-validate
Non-echo rating	rlens non-echo-rate
Non-echo analysis	rlens non-echo-analyse
Small-sample stability	rlens small-sample
Figures	rlens figures
Reproducibility manifest	rlens manifest

Two rubrics are used, each hashed and salted separately, and the software refuses to rate the panel under the non-echo rubric unless that rubric’s own copy control has passed *for the exact rubric text* being used. Artifact hashes, both protocol salts, all seeds and the audit verdicts are recorded in `final_reproducibility_manifest.json`. The unblinding key is deliberately absent from the repository and from this document’s source tree.

A The two rubrics, in full

Hashes establish immutability, not validity. Both system prompts are reproduced verbatim so a reader can judge what the raters were actually asked. Each was presented as the system message; the user message carried the prompt with the readout position marked `[[>>token<<]]`, the position index, and the three candidate arms as ranked token lists labelled A, B and C.

Standard rubric (salt coherence-v2-2026-08-26)

You are rating three anonymous readouts of the SAME internal state of a language model, taken at the SAME highlighted position in the prompt below. Each candidate was produced by a different decoding method. You do not know which is which, and you must not speculate about their identity.

Treat each readout as a BAG OF TOKEN-LEVEL CONCEPTS, not as a grammatical sentence. Token fragments may jointly express a meaningful word. Do NOT penalise valid non-English words, transliterations, or non-Latin scripts merely for not being English. Whitespace or punctuation may be contextually appropriate - judge it in context.

Score each candidate on three scales.

CONTEXTUAL COHERENCE (0-4) - the primary measure. Does the readout form a meaningful interpretation of the model's state at the highlighted position? 0: Predominantly malformed, uninterpretable, or semantically unrelated. 1: Mostly incoherent; at most one weak or plausibly accidental contextual connection. 2: Partially coherent; at least one clear relevant concept or theme, but substantial noise or ambiguity remains. 3: Mostly coherent; multiple tokens support a meaningful theme related to the prompt at the highlighted position. 4: Strongly coherent; a clear, semantically unified, contextually appropriate representation. Prompt copying alone CAN be coherent, but must not automatically earn a high score. Score whether the readout forms a meaningful contextual representation.

LEXICAL INTEGRITY (0-2) - are the outputs interpretable units, regardless of relevance? 0: Mostly malformed fragments, special tokens, punctuation runs, or meaningless repetitions. 1: A mixture of interpretable lexical items and malformed/noisy items. 2: Mostly recognisable words or meaningful token fragments.

PROMPT ECHO (0-2) - how much is direct repetition of visible input? Neither good nor bad. 0: Little or no direct copying of nearby prompt tokens. 1: Some copying, alongside additional content. 2: Dominated by direct prompt copying or trivial lexical variants.

Also return the contextual winner: "A", "B", "C", or "tie", and for each candidate an evidence statement of at most 25 words. Do not emit hidden reasoning.

Return STRICT JSON only, exactly this shape: {"A": {"contextual_coherence": 0, "lexical_integrity": 0, "prompt_echo": 0, "evidence": ""}, "B": {"contextual_coherence": 0, "lexical_integrity": 0, "prompt_echo": 0, "evidence": ""}, "C": {"contextual_coherence": 0, "lexical_integrity": 0, "prompt_echo": 0, "evidence": ""}, "contextual_winner": "A|B|C|tie"}

Non-echo rubric (salt non-echo-2026-08-27)

You are rating three anonymous readouts of the SAME internal state of a language model, taken at the SAME highlighted position in the prompt below. Each candidate was produced by a different decoding method. You do not know which is which, and you must not speculate about their identity.

Treat each readout as a BAG OF TOKEN-LEVEL CONCEPTS, not as a grammatical sentence. Token fragments may jointly express a meaningful word. Do NOT penalise valid non-English words, transliterations, or non-Latin scripts merely for not being English.

THE CENTRAL INSTRUCTION. Some readouts repeat words that already appear in the prompt. Before scoring, mentally DISCARD every token that merely repeats a word visible in the prompt above, including trivial variants (case, leading space, plural, a shared word stem). Score only what REMAINS after discarding those. A readout that is entirely prompt repetition has nothing remaining and scores 0, however fluent it looks.

NON-ECHO COHERENCE (0-4) - the primary measure. Of the tokens that are NOT copied from the prompt, do they form a meaningful interpretation of the model's state at the highlighted position? 0: Nothing remains after discarding copied tokens, or what remains is malformed, uninterpretable, or semantically unrelated. 1: What remains is mostly incoherent; at most one weak or plausibly accidental contextual connection. 2: What remains is partially coherent; at least one clear relevant concept or theme, but substantial noise or ambiguity. 3: What remains is mostly coherent; multiple non-copied tokens support a meaningful theme related to the prompt at the highlighted position. 4: What remains is strongly coherent; a clear, semantically unified, contextually appropriate representation built from non-copied tokens.

RESIDUAL SUBSTANCE (0-2) - how much non-copied material was there to judge? 0: Essentially nothing; the readout is dominated by prompt repetition. 1: Some non-copied material alongside substantial repetition. 2: Mostly non-copied material. Report this honestly. It is a description of the readout, not a quality score, and a low value does NOT justify raising or lowering non-echo coherence.

Also return the non-echo winner: "A", "B", "C", or "tie", and for each candidate an evidence statement of at most 25 words naming which tokens you discarded as copied. Do not emit hidden reasoning.

Return STRICT JSON only, exactly this shape: {"A": {"non_echo_coherence": 0, "residual_substance": 0, "evidence": ""}, "B": {"non_echo_coherence": 0, "residual_substance": 0, "evidence": ""}, "C": {"non_echo_coherence": 0, "residual_substance": 0, "evidence": ""}, "non_echo_winner": "A|B|C|tie"}

References

1. C. Blank, A. Bhatia and N. Nanda. *R-Lens: Making J-Lens More Faithful on Early Layers*. LessWrong / AlignmentForum, August 2026. The source of the coherence claim under evaluation. No scorer was released with it.
2. The J-Lens method and its released artifacts, as distributed in the `jlens` library (commit 581d398) and the `camilablank/workspace-lenses` artifact repository (commit d740106d).
3. S. Bach et al. *On Pixel-Wise Explanations for Non-Linear Classifier Decisions by Layer-Wise Relevance Propagation*. PLoS ONE, 2015. The relevance-propagation family whose backward rules R-Lens adapts.
4. L. Zheng et al. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. NeurIPS Datasets and Benchmarks, 2023. Position bias and model-dependent instability in LLM raters; the reason arm order is permuted per (cell, rater) here and position bias is an admission gate.
5. J. Montgomery, B. Nyhan and M. Torres. *How Conditioning on Post-Treatment Variables Can Ruin Your Experiment and What to Do About It*. American Journal of Political Science, 2018. The basis for treating the equal-echo restriction (§5.4) as descriptive rather than as an estimate of a direct effect.

6. A. C. Cameron, J. B. Gelbach and D. L. Miller. *Bootstrap-Based Improvements for Inference with Clustered Errors*. Review of Economics and Statistics, 2008. Cluster-level resampling, applied here at the prompt.
7. P. Good. *Permutation, Parametric and Bootstrap Tests of Hypotheses*. Springer, 3rd ed., 2005. Sign-flip permutation inference for paired designs.
8. J. Cohen. *Weighted Kappa: Nominal Scale Agreement with Provision for Scaled Disagreement or Partial Credit*. Psychological Bulletin, 1968. The quadratic-weighted κ reported in §5.3.
9. S. Holm. *A Simple Sequentially Rejective Multiple Test Procedure*. Scandinavian Journal of Statistics, 1979. Multiplicity correction within model for the depth and set breakdowns.
10. C. J. Clopper and E. S. Pearson. *The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial*. Biometrika, 1934. Exact intervals on the order-flip rates in §5.2.

Artifacts. Repository `SamanyuB910/CAMRBIA-CapStone`, branch `coherence-branch`. Every figure and table in this document is produced by a command listed in §13 operating on frozen artifacts. Panel hash, both rubric hashes, both protocol salts, all seeds, the exact judge model identifiers and the SHA-256 of every artifact are recorded in `final_reproducibility_manifest.json`.